

# NAQEEB ARSHAD

## Mechanical Design Engineer

Mechanical design, CAD modelling, CNC manufacturing, fabrication, and product development.

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|              |                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------|
| CURRENT ROLE | Design Engineer - Tech Valley, Islamabad                                                                  |
| Address      | Rawalpindi, Pakistan                                                                                      |
| EMAIL        | naqeebarshad402@gmail.com                                                                                 |
| PHONE        | +92 310 5038342                                                                                           |
| LINKEDIN     | <a href="https://www.linkedin.com/in/naqeeb-arshad-455120373">linkedin.com/in/naqeeb-arshad-455120373</a> |



# Design. Validate. Manufacture.

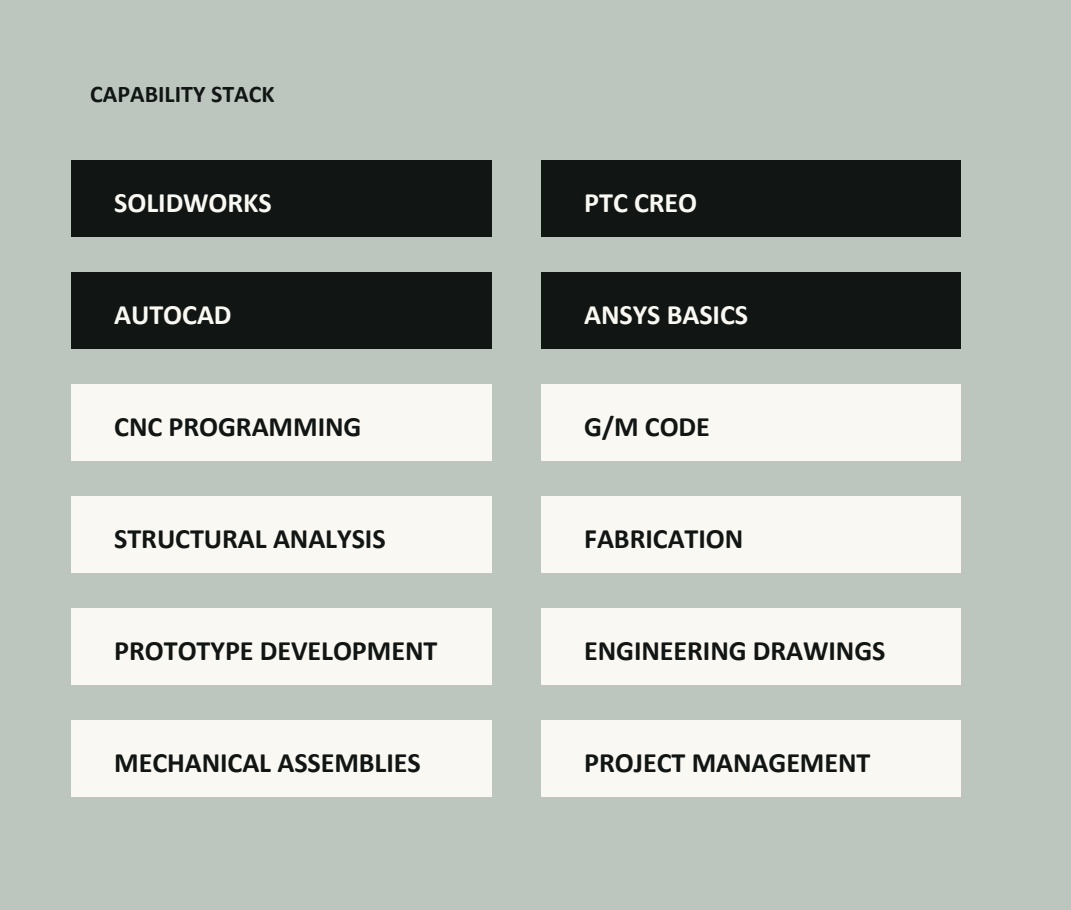
Mechanical engineering experience across CAD modelling, fabrication, machining, and product development.

Precise enough for engineering drawings.

*Practical enough for manufacturing.*

I transform engineering concepts into practical, manufacturable solutions through mechanical design, CAD modelling, engineering drawings, machining preparation, and prototype development.

|              |                                                           |
|--------------|-----------------------------------------------------------|
| CURRENT ROLE | Design Engineer - Tech Valley, Islamabad                  |
| EDUCATION    | BS Mechanical Engineering  3.18   HITEC University Taxila |
| LOCATION     | Islamabad, Pakistan                                       |
| EMAIL        | naqeebarshad402@gmail.com                                 |
| PHONE        | +92 310 5038342                                           |



# Professional Background

Verified roles, education, certifications, and achievements from the supplied portfolio and resume.

PROFESSIONAL EXPERIENCE

|              |                                                                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CURRENT      | <div>Design Engineer</div> <div>Tech Valley, Islamabad</div> <div>SolidWorks design of drone-handling, rotary-cleaning, and vehicle-mounted systems.</div>  |
| JUL-SEP 2025 | <div>Intern</div> <div>NUST College of EME (Robot Maker Lab)</div> <div>Eight-week internship in design, manufacturing, CNC cutting, and fabrication.</div> |
| APR-JUN 2022 | <div>Assistant Engineer</div> <div>Electrosolz, Islamabad</div> <div>Product design and manufacturing work involving 3D printers.</div>                     |
| INDEPENDENT  | <div>Freelance CAD Designer</div> <div>Remote</div> <div>AutoCAD house plans and SolidWorks mechanical components for client requirements.</div>            |

EDUCATION

2022-2026

BS Mechanical Engineering

HITEC University Taxila | CGPA 3.18 / 4.00

2019-2021

DAE Mechanical

KRL Institute of Technology Kahuta | 93%

CERTIFICATIONS & ACHIEVEMENTS

+ NINVEST - 2nd International Additive Manufacturing and Vacuum Technology Conference

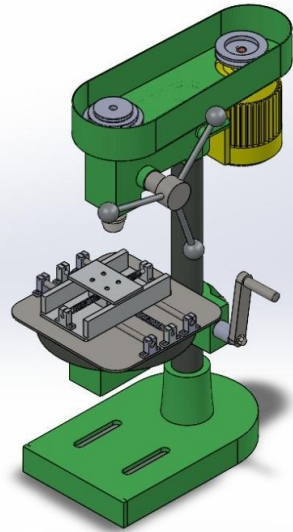
+ NAVTTC - Six-month AutoCAD Diploma (2022)

+ Technical Head - Robo Fiesta 8.0 | Two-time HI-Robo Tec paper-plane winner

+ Letter of Appreciation - 11th Open House 2025, HITEC University Taxila

# Friction-Based Metal Printing Machine (AFSD)

Manual desktop Additive Friction Stir Deposition machine for AA6061 aluminium, completed at HITEC University Taxila.



+46.2%

STRENGTH IMPROVEMENT

+4%

HARDNESS VS. CAST-WELDED AA6061

## FINAL YEAR PROJECT

### Manual desktop AFSD machine

The complete mechanical design was developed in SolidWorks. Primary load-bearing components were checked through static structural and vibration analysis before fabrication.

- + Complete SolidWorks assembly
- + Static structural analysis
- + Vibration and frequency analysis
- + Successful AFSD deposition in Pakistan

# Verified AFSD Results

Results reported in the reference portfolio after design validation, fabrication, deposition, and material testing.

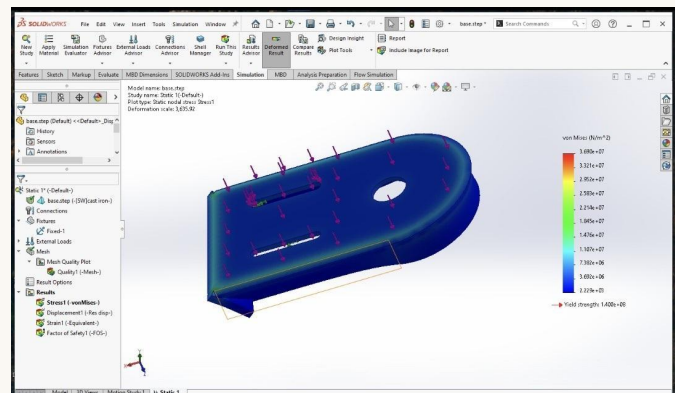
|                                                                                                                              |                                                                                                                 |                                                                                                                            |                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <div>01 / VERIFIED</div> <div>1st in Pakistan</div> <div>Successful metal deposition using the developed AFSD process.</div> | <div>02 / VERIFIED</div> <div>+4% hardness</div> <div>Improvement over conventionally cast-welded AA6061.</div> | <div>03 / VERIFIED</div> <div>+46.2% strength</div> <div>Compression-strength improvement in the deposited specimen.</div> | <div>04 / VERIFIED</div> <div>24.84 Hz</div> <div>First natural frequency confirmed by modal analysis.</div> |
|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|

PROJECT VERIFICATION

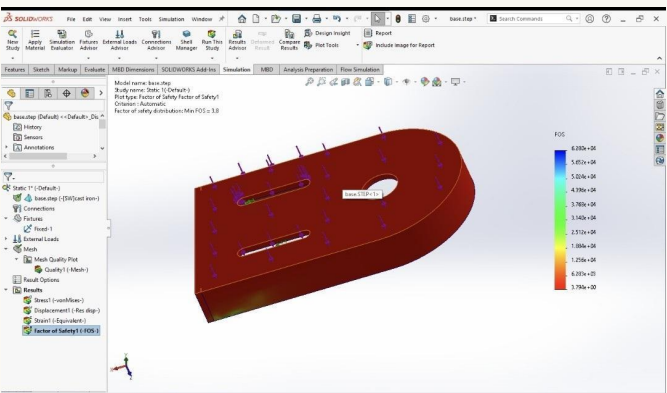
|                                                                                                              |                                                                                                      |                                                                                                      |                                                                                                            |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <div>01</div> <div>SolidWorks Design</div> <div>Complete machine assembly and load-bearing components.</div> | <div>02</div> <div>Structural Analysis</div> <div>Static structural checks before fabrication.</div> | <div>03</div> <div>Vibration Analysis</div> <div>First natural frequency verified at 24.84 Hz.</div> | <div>04</div> <div>Material Testing</div> <div>Compression and hardness testing of deposited AA6061.</div> |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|

# Base Plate - Structural & Vibration Analysis

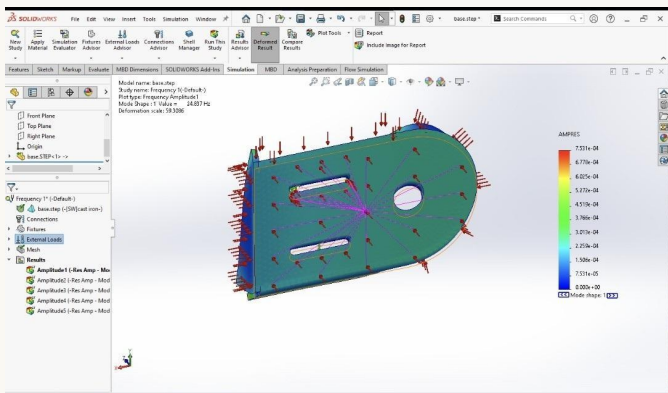
Static structural and modal analysis verified operating-load capacity and frequency separation before fabrication.



VON MISES STRESS



FACTOR OF SAFETY



MODAL FREQUENCY

**OBJECTIVE**

Verify the base under operating loads and locate critical stress.

**METHOD**

Static structural study plus first-mode frequency evaluation.

**RESULT**

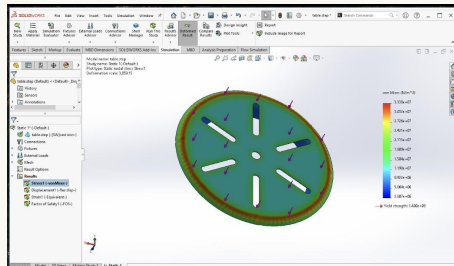
36.9 MPa stress vs. 140 MPa yield; FOS 3.8; first mode 24.84 Hz.

# Table & Support Column - Structural Analysis

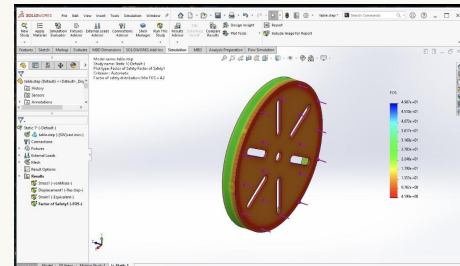
The rotating table plate and vertical support column were verified as critical load-path components of the AFSD machine.

## 02 / SIMULATION PROJECT

### TABLE PLATE



STRESS



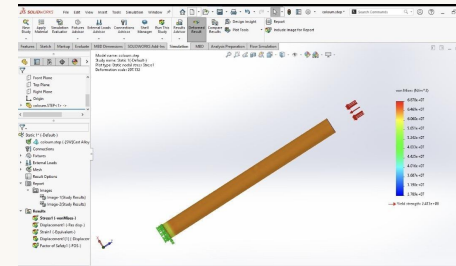
SAFETY FACTOR

#### VERIFIED RESULT

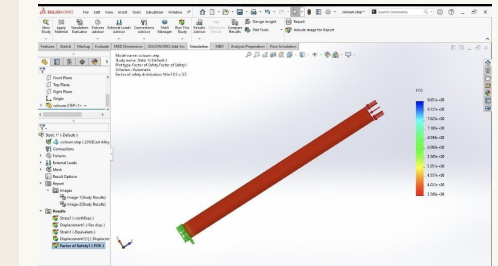
33.4 MPa maximum stress vs. 140 MPa yield. Validated FOS: 4.2.

## 03 / SIMULATION PROJECT

### SUPPORT COLUMN



STRESS



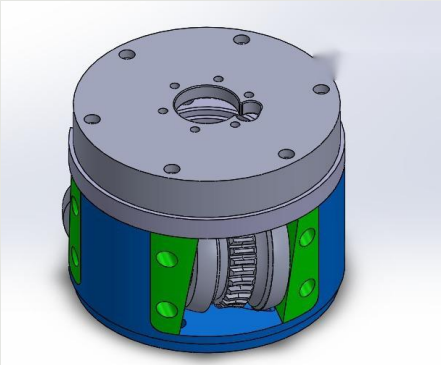
SAFETY FACTOR

#### VERIFIED RESULT

68.8 MPa maximum stress vs. 241 MPa yield. Validated FOS: 3.5.

# Defense & Advanced Systems Design at Tech Valley Islamabad

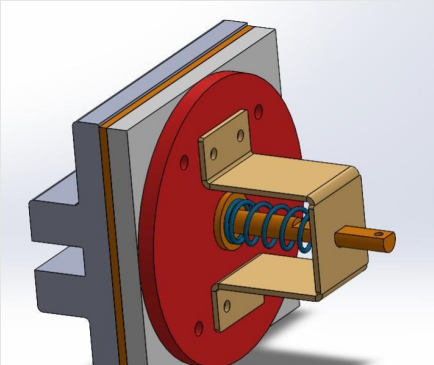
Mechanical assemblies for drone handling, rotary cleaning equipment, and vehicle-mounted structural systems.



01 / SOLIDWORKS

## Tank Barrel Cleaner - Assembly Design

|                 |                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| PURPOSE         | Rotary cleaning of internal tank-barrel surfaces.                                                            |
| DESIGN WORK     | Designed the cylindrical housing, internal drive gear train, and top mounting flange with bolt hole pattern. |
| PROJECT CONTEXT | Design Engineer   Tech Valley, Islamabad                                                                     |



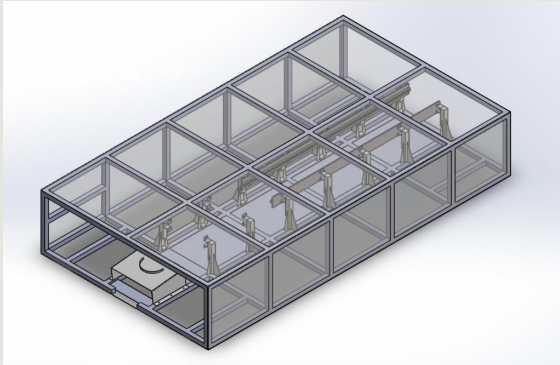
02 / SOLIDWORKS

## Drone Locking System - Mechanism Design

|                 |                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------|
| PURPOSE         | Secure a drone payload in position.                                                     |
| DESIGN WORK     | Designed the locking arms, guide pin, compression spring assembly, and mounting flange. |
| PROJECT CONTEXT | Design Engineer   Tech Valley, Islamabad                                                |

# Defense & Advanced Systems Design at Tech Valley Islamabad

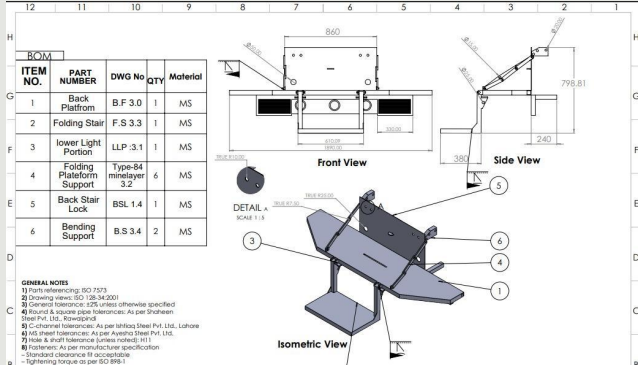
Mechanical assemblies for drone handling, rotary cleaning equipment, and vehicle-mounted structural systems.



03 / SOLIDWORKS

## Drone Launcher - Multi-Cell Deployment Rack

|                 |                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------|
| PURPOSE         | Sequential deployment from a modular multi-cell frame.                                         |
| DESIGN WORK     | Designed the deployment station grid and sliding drive mechanism for sequential drone release. |
| PROJECT CONTEXT | Design Engineer   Tech Valley, Islamabad                                                       |



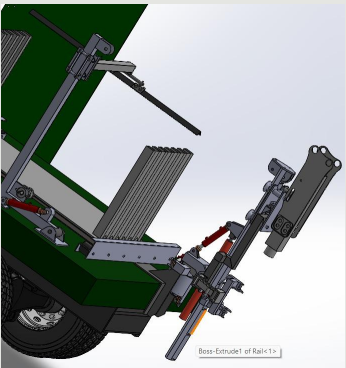
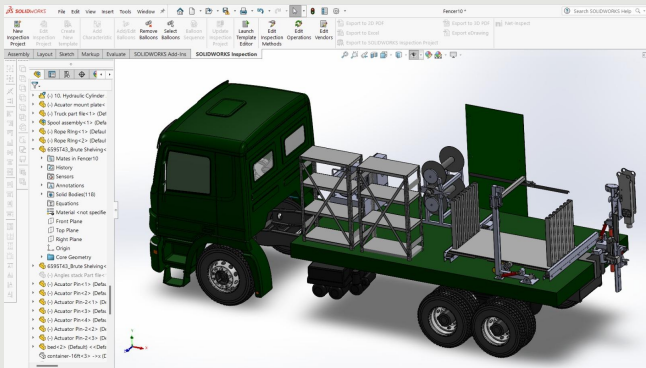
04 / SOLIDWORKS

## Folding Stair & Platform Mechanism - Vehicle-Mounted Assembly

|                 |                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------|
| PURPOSE         | Vehicle mounted folding access system.                                                         |
| DESIGN WORK     | Prepared the mechanism, supports, locks, views, BOM, tolerance notes, and fabrication details. |
| PROJECT CONTEXT | Design Engineer   Tech Valley, Islamabad                                                       |

# Defense & Advanced Systems Design at Tech Valley Islamabad

Truck-mounted mechanized system for automated perimeter fence installation around army minefields.



05 / SOLIDWORKS

## Mine-Fencing Rig - Truck-Mounted Assembly

|                 |                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| PURPOSE         | Mechanized fence-laying system replacing manual minefield perimeter marking.                                                 |
| DESIGN WORK     | Designed the full truck-mounted assembly integrating storage racks, fence-roll feed reels, and the rear picket driving rail. |
| PROJECT CONTEXT | Design Engineer   Tech Valley, Islamabad                                                                                     |

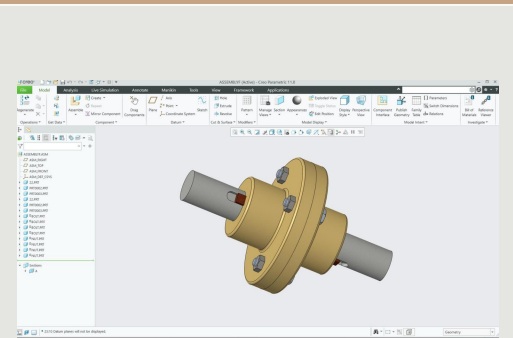
06 / SOLIDWORKS

## Automatic Hammer & Rail Mechanism - Fence-Post Driver

|                 |                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------|
| PURPOSE         | Automated picket-driving unit that replaces manual hammering in the field.                                  |
| DESIGN WORK     | Designed the hydraulic hammer, guide rail, and actuator-pin linkage that drive fence posts into the ground. |
| PROJECT CONTEXT | Design Engineer   Tech Valley, Islamabad                                                                    |

# PTC Creo Design & CNC Programming

Internship work covering part and assembly modelling, CNC leaser cutting and manufacturing setup, toolpath generation, and NC code.

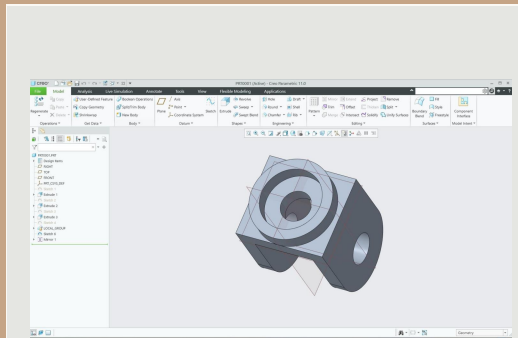


## ASSEMBLY DESIGN

### Flange Coupling Assembly Design with Fastener Pattern

Bolt-hole fastener pattern and shaft interface modelled in PTC Creo.

01 / NUST INTERNSHIP

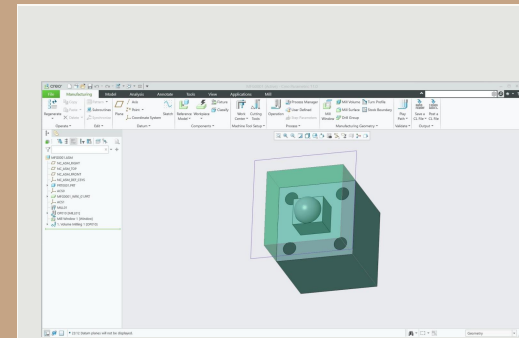


## PART DESIGN

### Profile-Milled Component Part Design

Stepped bores and a chamfered profile prepared for CNC machining.

02 / NUST INTERNSHIP



## MANUFACTURING

### CNC Manufacturing Setup Volume Milling Operation

Volume-milling setup and toolpath generation in PTC Creo Manufacturing.

03 / NUST INTERNSHIP

```
%  
(DATE:09/04/25 TIME:12:15:46)  
G98G80G90G49G17  
( / MFG0001)  
T1M6  
S200M3  
G0X50.Y0.  
G43Z5.H1  
G81X50.Y0.Z-18.906R1.F100.  
G80  
G0Z5.  
M5  
T30M6  
M30  
%
```

## CNC PROGRAMMING

### Generated NC Code (G-code / M-code)

NC program generated from the PTC Creo manufacturing operation.

04 / NUST INTERNSHIP

01 MODEL

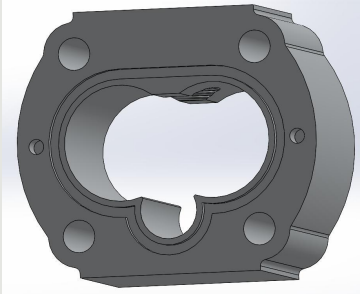
02 TOOLPATH

03 G/M CODE

04 MACHINE

# Component & Assembly Design Work

SolidWorks work completed during the NUST internship and the AFSD final-year project.

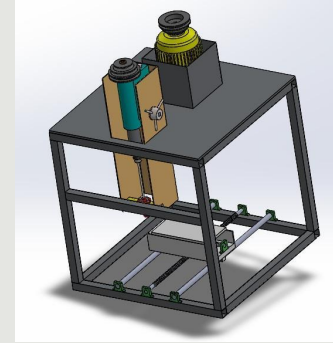


## SOLIDWORKS COMPONENT DESIGN

### Housing Design

Head housing with motor mount interface, and bolt-hole pattern.

NUST College of EME | Internship

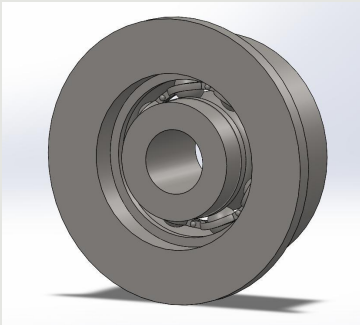


## SOLIDWORKS ASSEMBLY DESIGN

### AFSD Machine Frame & Table Assembly Design

Complete structural frame and worktable with motor mounting, linear guides, and base support.

HITEC University Taxila | Final Year Project

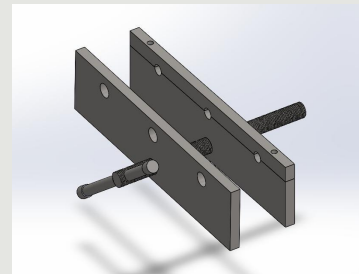


## SOLIDWORKS POWER TRANSMISSION

### Bearing & Pulley Housing Design

Bearing and pulley housing assembly sized to standard bearing tolerances.

NUST College of EME | Internship



## SOLIDWORKS FIXTURE DESIGN

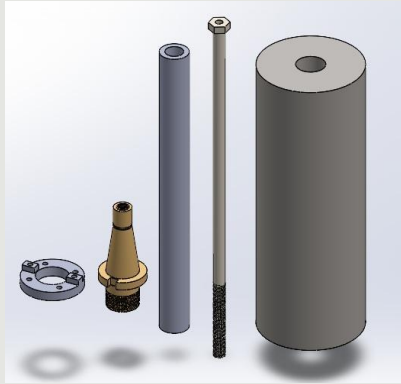
### Bolted Clamp Plate Fixture

Bolted clamp-plate fixture with an adjustable bolt hole pattern for workpiece alignment.

NUST College of EME | Internship

# Component & Assembly Design Work

University and internship projects are identified separately with their verified project context.

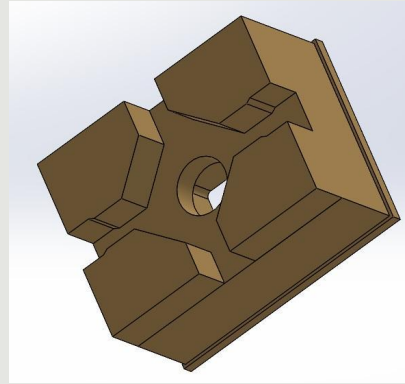


SOLIDWORKS UNIVERSITY PROJECT

## Vertical Milling Machine Head

Vertical milling machine head and related components modelled for FYP understanding.

HITEC University Taxila | Final Year Project

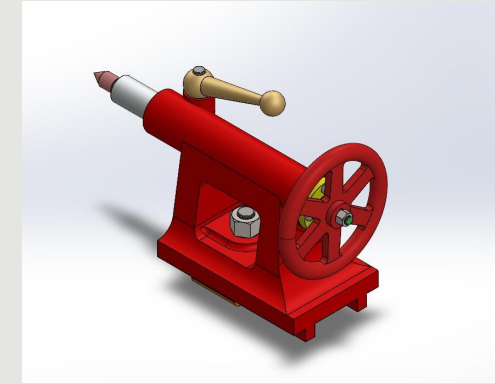


SOLIDWORKS MACHINED COMPONENT

## Machined Bracket Component

Mounting bracket with stepped faces and a central bore, prepared for VMC machining.

NUST College of EME | Internship



SOLIDWORKS SEMESTER PROJECT

## Tail stock of lathe machine

Manual hand tail stock assembly modelled for lathe machine operations.

HITEC University Taxila | Semester Project

# AutoCAD Design Projects

Residential design work completed while learning AutoCAD through the NAVTTC six-month diploma course.

## COURSE PROJECT

## Residential 3D House Plan and Layout

Completed industry-based architectural design work in AutoCAD, including a residential 3D house plan and layout.

### SOFTWARE

AutoCAD

### COURSE

NAVTTC 6-month AutoCAD Diploma

### PROJECT TYPE

Residential 3D house plan and layout



# Additional Academic Projects

Academic projects on Solid Works and Auto Cad.

01

HEAT TRANSFER

## Fins Design Verification under Forced Convection

Verified circular and rectangular fin designs by comparing actual lengths with optimized theoretical lengths for maximum heat-transfer performance.

HITEC UNIVERSITY TAXILA

02

RENEWABLE ENERGY SYSTEM

## Wind-Driven Water Pump

Designed a wind-driven mechanical system for water pumping in agricultural areas using renewable energy instead of conventional power.

HITEC UNIVERSITY TAXILA

03

VIBRATION ANALYSIS

## Power Transmission Mechanism for Vibration Analysis

Designed and developed a mechanical power-transmission mechanism for demonstration in vibration-analysis experiments.

HITEC UNIVERSITY TAXILA

AVAILABLE FOR MECHANICAL DESIGN OPPORTUNITIES

Let's turn a complex  
requirement into a  
*working design.*

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**NAQEEB ARSHAD**

MECHANICAL DESIGN ENGINEER / RAWALPINDI, PAKISTAN

